

ARCHITECTURE & DESIGN DOCUMENT

Annex

Extra desktop, no extra hardware.

A Rust system that turns a Windows PC into a wireless **extended** monitor for a Mac. No HDMI, no cables.

Rust

macOS host

Windows / browser client

WebRTC

ScreenCaptureKit

VideoToolbox

CGVirtualDisplay

LAN / Wi-Fi

Project name: Annex · **Version:** 1.0 (design) · **Date:** 24 July 2026

Goal: Mac is the primary computer; a second laptop/PC becomes a real, extended second desktop over the local network.

Audience: the developer(s) building it. Assumes general programming ability; explains all macOS/Rust specifics.

Table of Contents

1. Executive Summary

What we are building

The core insight (two halves)

2. Goals, Non-Goals & Success Criteria

3. Technical Background & Key Constraints

4. High-Level Architecture

System context diagram

Process model

End-to-end data flow

5. Component Design

virtual-display

capture

encoder

transport & signaling

web client

input (phase 2)

core / host app

6. Workspace & Crate Layout

7. Concurrency & Threading Model

8. Sequence Diagrams

9. Key Data Structures & APIs

10. Performance & Latency Budget

11. Security, Permissions & Privacy

12. Dependencies (Crates) & Build

13. Risks & Mitigations

14. Roadmap & Milestones

15. Testing & Verification Strategy

16. Appendix A: Reverse-engineered CGVirtualDisplay header

17. Appendix B: References

18. Appendix C: Glossary

1. Executive Summary

1.1 What we are building

Annex is a two-part application. On the Mac (the *host*) a Rust program creates a **virtual monitor** that macOS treats as a genuine second display, and you can drag windows onto it exactly as if a physical monitor were plugged in. The Rust program captures that virtual monitor's pixels, encodes them to video, and streams them over Wi-Fi/LAN to a *client* running on the Windows laptop. The client renders the video full-screen, so the Windows laptop **becomes** the second monitor. No HDMI cable, no capture card, just the two machines on the same network.

Version 1's client is an ordinary **web browser** (zero install on Windows); a native Rust client is a later milestone. Optionally, the client can send mouse/keyboard input back to the Mac so the second screen is interactive, not just a viewer.

Why this is fundamentally possible: every piece already exists in shipping software: **DeskPad** and **BetterDummy** prove the virtual-display trick; **Deskreen** proves the browser-streaming trick. Annex combines both, in one Rust codebase, as a true *extended* (not mirrored) display.

1.2 The core insight: this is two separate problems

The single most important thing to understand before writing any code is that the project is **two very different sub-systems** bolted together. They share almost no techniques and differ enormously in difficulty.

Half	Job	Mechanism	Difficulty
Half 1: Virtual display	Make macOS believe a monitor is attached, so windows can live there	Private, undocumented CGVirtualDisplay APIs inside CoreGraphics	Hard (private API)
Half 2: Capture + stream	Grab that display's pixels, encode, send to Windows, render	ScreenCaptureKit → VideoToolbox → WebRTC → browser	Well-trodden (standard video streaming)

If we build only Half 2, we have "our own Deskreen" that mirrors the main screen and relies on a third-party tool (BetterDisplay) for the extend. If we build **both**, Annex is a complete, self-contained product. The roadmap (§14) deliberately sequences Half 2 first (it is de-riskable and demoable early) and folds in Half 1 once the streaming pipeline works end-to-end.

2. Goals, Non-Goals & Success Criteria

2.1 Goals (v1)

- **Mac as host:** the Mac is the primary computer whose desktop is extended.
- **True extend, not mirror:** the Windows screen is a *new* desktop area, not a copy of the Mac's screen.
- **Wireless over LAN:** both machines on the same Wi-Fi/router; no HDMI, no capture hardware.
- **Zero-install client (v1):** Windows just opens a URL in a browser.
- **One language:** the host and (later) native client are both Rust.
- **Usable latency:** smooth for reference material, docs, dashboards, video playback; target < 60 ms glass-to-glass on good Wi-Fi.
- **Configurable resolution & HiDPI** for the virtual display.

2.2 Non-goals (explicitly out of scope for v1)

- **Internet / NAT traversal.** LAN only. No STUN/TURN servers, no cloud relay.
- **Mac App Store distribution.** The private virtual-display API forbids it; we ship as a signed local app.
- **Audio** forwarding.
- **The reverse direction** (Windows host → Mac client).
- **Interactive input** is a *phase-2* goal, not v1.
- **Multi-client / multi-monitor fan-out**, designed-for but not built in v1.

2.3 Success criteria (definition of done for v1)

1. Launch host on Mac → a new display appears in **System Settings → Displays**.
2. Open the host's URL in a Windows browser on the same network → the virtual display's contents appear full-screen.
3. Drag a window from the Mac's main screen onto the virtual display → it shows up on the Windows laptop.
4. Motion (scrolling, a playing video) is smooth (≥ 30 fps) with sub-100 ms perceived lag on Wi-Fi.
5. Closing the host cleanly removes the virtual display (no ghost monitors left behind).

3. Technical Background & Key Constraints

3.1 The macOS media stack we depend on

Concern	API / Framework	Public?	Notes
Create virtual monitor	<code>CGVirtualDisplay*</code> (CoreGraphics)	❌ Private	4 undocumented Obj-C classes. No kext, no DriverKit, no root, no SIP-disable.
Capture a display's pixels	ScreenCaptureKit (<code>SCStream</code>)	✅ Public	macOS 12.3+. GPU-accelerated, delivers <code>CVPixelBuffer</code> . Requires Screen Recording permission.
Hardware video encode	VideoToolbox (<code>VTCompressionSession</code>)	✅ Public	C API. H.264 & HEVC on the media engine. Accepts <code>CVPixelBuffer</code> directly (near zero-copy).
Real-time transport	WebRTC (webrtc-rs)	✅ Pure Rust	No native libwebrtc build. Congestion control, packet loss recovery, browser-compatible.
Input injection (phase 2)	<code>CGEvent</code> (CoreGraphics)	✅ Public	Requires Accessibility permission. Maps client input → Mac events.

The one true difficulty: `CGVirtualDisplay` is undocumented. It is stable enough that DeskPad/BetterDummy rely on it across many macOS releases, but Apple can change its shape between versions. We isolate **all** private-API contact inside a single crate so breakage is contained to one file (§5.1, §13).

3.2 Why a virtual display is required at all

Screen-sharing tools can only capture pixels that macOS is *already rendering somewhere*. To get a *new* desktop area (extend), some display must exist for the windows to occupy. A physical monitor does this via HDMI/EDID; a "dummy plug" fakes it in hardware; `CGVirtualDisplay` fakes it in **software**. Once the fake display exists, capturing it and streaming it produces a genuine extended second monitor.

3.3 Constraints that shape the whole design

- **Run loop affinity:** ScreenCaptureKit, AppKit (tray UI), and the virtual-display callbacks want a thread with a Core Foundation run loop, i.e. the **main thread**. WebRTC/async must live on separate (tokio) threads and communicate via channels.
- **Zero-copy path:** ScreenCaptureKit outputs `CVPixelBuffer` ; VideoToolbox consumes `CVPixelBuffer` . Keeping frames on the GPU from capture → encode avoids expensive CPU copies.
- **Browser compatibility:** because v1's client is a browser, the encoder must emit a browser-decodable codec/profile. **H.264 (Constrained Baseline / Main, Annex-B)** is the safe universal choice; HEVC is an optimization for capable clients.
- **LAN-only:** WebRTC ICE only needs **host candidates**; no STUN/TURN. Simplifies signaling enormously.

4. High-Level Architecture

4.1 System context

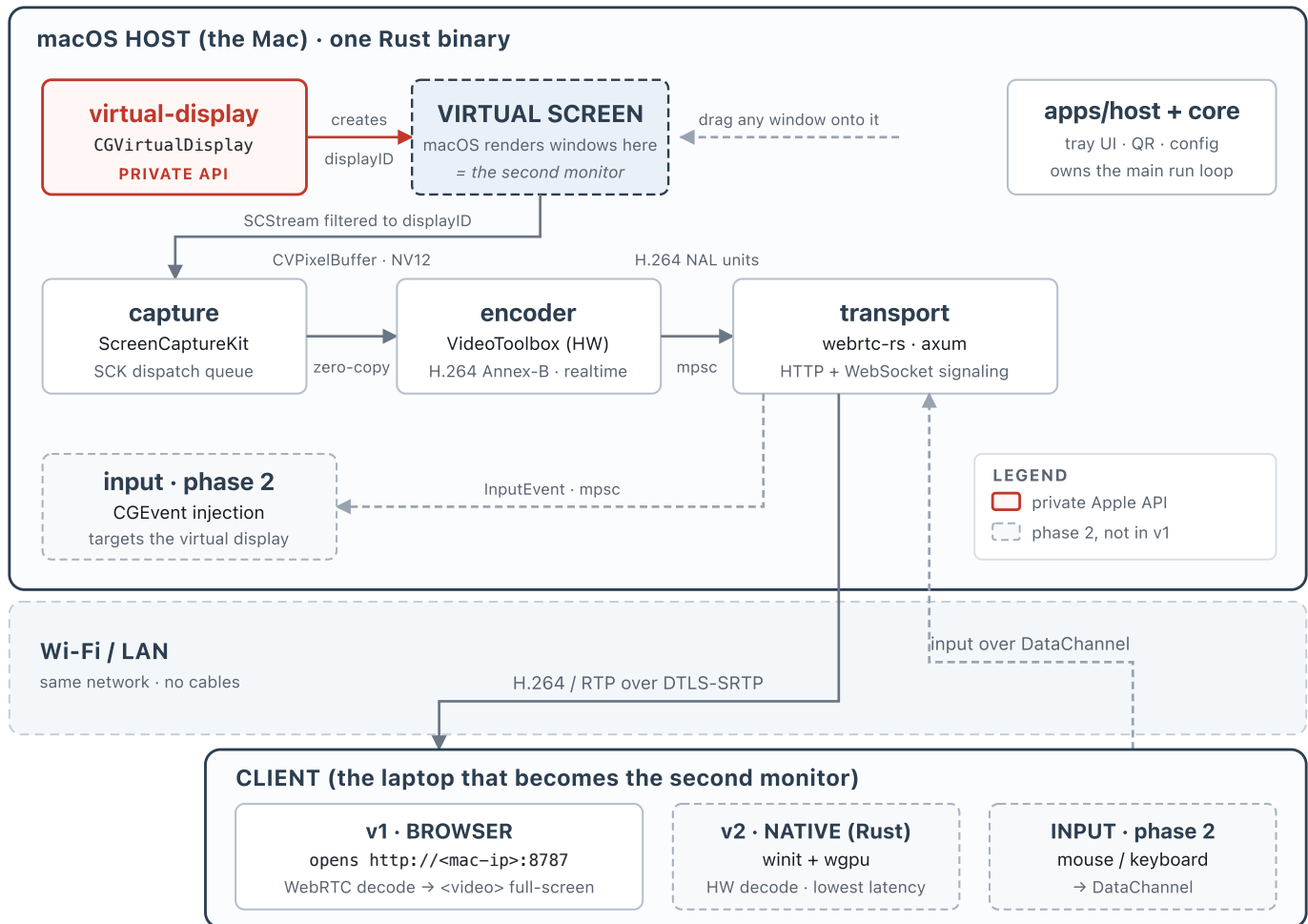


Figure 1. System context. Red border marks the single crate that touches private Apple APIs; dashed boxes are phase 2. The `displayID` produced by `virtual-display` is the handle `capture` uses to open its `SCStream`.

4.2 Process model

- **Host process (macOS):** a single native Rust binary, presented as a menu-bar/tray app. It owns the virtual display, the capture+encode pipeline, an embedded HTTP server (serves the client page), a WebSocket signaling endpoint, and one WebRTC peer connection per connected client.
- **Client (v1 = browser):** a static HTML/JS page (served by the host) that establishes a WebRTC session and paints the incoming video track. Nothing to install on Windows.
- **Client (v2 = native Rust):** a `winit + wgpu` window that decodes the stream and renders it, for lower latency and a cleaner full-screen experience. Same signaling protocol.

4.3 End-to-end data flow (steady state)

1. macOS composites the windows sitting on the virtual display into its framebuffer.

2. **ScreenCaptureKit** delivers each new frame as a `CVPixelBuffer` on its dispatch queue.
3. The frame is handed (zero-copy) to **VideoToolbox**, which emits an H.264 access unit (NAL units).
4. Encoded samples cross an **mpsc channel** into the tokio world.
5. **webrtc-rs** packetizes them into RTP and sends over the peer connection.
6. The browser's WebRTC stack depacketizes, decodes (hardware), and renders to a `<video>` element.
7. (*phase 2*) the client captures pointer/key events, sends them on a **DataChannel**; the host maps coordinates into the virtual display and injects them with `CGEvent` .

5. Component Design

Each component is a Rust crate with a narrow, testable surface. Below: responsibility, public API sketch, and the key implementation notes/gotchas.

5.1 `virtual-display` : the private-API firewall

Responsibility: create/destroy a macOS virtual monitor and expose its `CGDirectDisplayID`. This is the **only** crate that touches private Apple APIs. It wraps the four Obj-C classes (`CGVirtualDisplayDescriptor`, `...Mode`, `...Settings`, `CGVirtualDisplay`) via `objc2` runtime class lookup, and uses RAI so dropping the handle removes the monitor.

```
pub struct VirtualDisplayConfig {
    pub name: String,
    pub width: u32, pub height: u32, pub refresh_hz: f64,
    pub hidpi: bool,           // true → Retina backing store
    pub size_mm: (f64, f64),   // physical size → controls default DPI
}

pub struct VirtualDisplay { /* retains the CGVirtualDisplay obj-c object */ }

impl VirtualDisplay {
    /// Looks up CGVirtualDisplay* classes at runtime, builds descriptor +
    /// settings, calls initWithDescriptor: / applySettings:.
    pub fn create(cfg: &VirtualDisplayConfig) -> Result<Self, VdError>;
    /// CGDirectDisplayID: hand this to the capture crate.
    pub fn display_id(&self) -> u32;
}

impl Drop for VirtualDisplay { /* release obj-c object → monitor disappears */ }
```

Gotchas: (1) the `CGVirtualDisplay` instance is the monitor, so it must be kept alive for the whole session; (2) the descriptor needs a valid **dispatch queue** (main queue works), since some macOS versions insist on it; (3) the `terminationHandler` is an Obj-C **block** (`block2 crate`), can be a no-op in v1; (4) copy the exact header from `DeskPad/BetterDummy`, do not transcribe field signatures from memory (see App. A).

5.2 capture : `ScreenCaptureKit` source

Responsibility: given a `display_id`, open an `SCStream` filtered to that display and deliver frames as `CVPixelBuffer`s to a callback/channel. Handles resolution, pixel format (BGRA or NV12), target FPS, and start/stop.

```
pub struct CaptureConfig { pub display_id: u32, pub fps: u32, pub pixel_format: PixFmt }
pub struct Capturer { /* holds SCStream + delegate */ }

impl Capturer {
    pub fn start(cfg: CaptureConfig, sink: FrameSink) -> Result<Self, CaptureError>;
    pub fn stop(self);
}
// FrameSink receives (CVPixelBuffer, presentation_timestamp) on SCK's queue.
```

Notes: resolve the `SCDisplay` whose `displayID` equals our virtual display via `SCShareableContent`. Request **NV12** output when feeding VideoToolbox (native encoder input, avoids a color convert). Set `minimumFrameInterval` for the target FPS. The delegate callback runs on a dispatch queue, so bridge into Rust and forward without blocking.

5.3 encoder : `VideoToolbox H.264/HEVC`

Responsibility: compress `CVPixelBuffer`s into an elementary stream of NAL units suitable for WebRTC. Configured for **real-time, low-latency** operation.

```
pub struct EncoderConfig {
    pub width:u32, pub height:u32, pub fps:u32,
    pub bitrate_kbps:u32, pub codec:Codec /* H264 | HEVC */,
    pub keyframe_interval:u32,
}
pub struct Encoder { /* VTCompressionSession */ }
impl Encoder {
    pub fn new(cfg:EncoderConfig, out:EncodedSink) -> Result<Self, EncError>;
    pub fn encode(&self, frame:&CVPixelBuffer, pts:Timestamp);
    pub fn request_keyframe(&self); // on new client / packet loss
    pub fn set_bitrate(&self, kbps:u32); // react to WebRTC congestion signals
}
```

Critical VideoToolbox settings for low latency: `RealTime = true`, `AllowFrameReordering = false` (no B-frames), `ProfileLevel = Baseline/Main AutoLevel`, `MaxKeyFrameInterval` tuned, and **Annex-B** output so WebRTC's H.264 packetizer can consume it. Emit SPS/PPS with each keyframe (in-band) for mid-stream client joins.

5.4 transport : `WebRTC` peer + signaling

Responsibility: (a) serve the client page over HTTP; (b) run a WebSocket signaling channel to exchange SDP + ICE; (c) manage a `RTCPeerConnection` per client with one video track (and a `DataChannel` for phase-2

input); (d) push encoded samples into the video track.

```
// signaling messages (JSON over WebSocket)
enum Signal { Offer(Sdp), Answer(Sdp), Ice(Candidate), Hello{token:String} }

pub struct Session { pc: RTCPeerConnection, video: Arc<TrackLocalStaticSample> }
impl Session {
    pub async fn accept(ws:WebSocket, cfg:&RtcConfig) -> Result<Session, RtcError>;
    pub async fn write_sample(&self, nal:&[u8], dur:Duration); // feed encoder output
    pub fn on_input(&self, cb: impl Fn(InputEvent));           // phase-2 DataChannel
}
```

Notes: since it is LAN-only, configure ICE with **no STUN/TURN**, since host candidates connect directly. Use `TrackLocalStaticSample` with the H.264 payload; `webrtc-rs` handles RTP + NACK + PLI. On a PLI (picture-loss indication) from the client, call `encoder.request_keyframe()`. Prefer an async HTTP framework (`axum`) for both the static page and the WebSocket upgrade on one port.

5.5 `web/client` : the browser client (v1)

Responsibility: a single static page. Opens a WebSocket to the host, performs the WebRTC handshake, attaches the remote track to a full-screen `<video autoplay playsinline>`, and (phase 2) forwards pointer/keyboard events. ~150 lines of vanilla JS; no framework needed.

```
const pc = new RTCPeerConnection({ iceServers: [] }); // LAN → no STUN/TURN
pc.ontrack = e => { videoEl.srcObject = e.streams[0]; };
const ws = new WebSocket(`ws://${location.host}/signal`);
// exchange offer/answer + ICE over ws ... then video.requestFullscreen()
```

5.6 `input` : CGEvent injection (phase 2)

Responsibility: translate `InputEvent` s from the client into macOS events targeted at the virtual display's coordinate space, via `CGEventCreateMouseEvent` / `...KeyboardEvent` + `CGEventPost`. Requires the user to grant **Accessibility** permission. Coordinate mapping must account for the virtual display's origin in the global display arrangement and HiDPI scale factor.

5.7 `core` + `apps/host` : orchestration & UI

core holds shared types (config, errors, the frame/sample structs, protocol enums) and the pipeline wiring. **apps/host** is the actual binary: it owns the main-thread run loop, builds a tray menu (`tray-icon` / `muda`), shows the connect URL + a QR code, starts the tokio runtime, and glues virtual-display → capture → encoder → transport together.

6. Workspace & Crate Layout

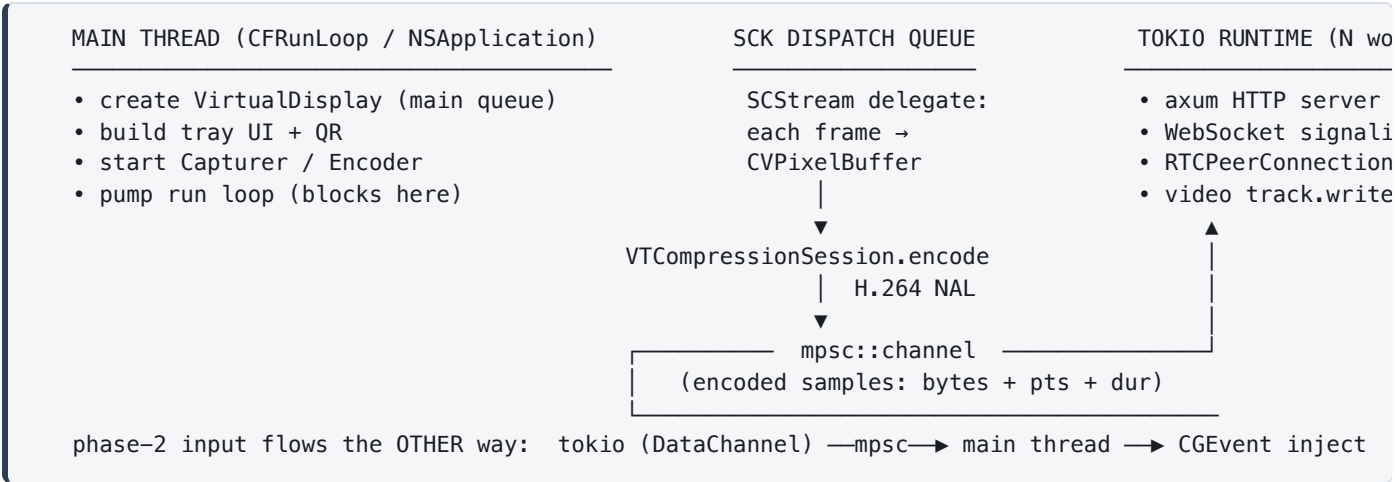
```
annex/
├─ Cargo.toml           # [workspace] members
├─ rust-toolchain.toml  # pin stable toolchain
├─ crates/
│   ├─ core/            # shared types, config, errors, pipeline glue
│   ├─ virtual-display/ # ← ONLY crate using private CGVirtualDisplay (objc2)
│   ├─ capture/         # ScreenCaptureKit → CVPixelBuffer frames
│   ├─ encoder/         # VideoToolbox → H.264 / HEVC NAL units
│   ├─ transport/       # webrtc-rs peer + axum HTTP/WebSocket signaling
│   └─ input/           # (phase 2) CGEvent injection
├─ apps/
│   ├─ host/            # macOS binary: tray UI, run loop, wires it all
│   └─ client-native/   # (phase 2) winit + wgpu + webrtc-rs client
└─ web/
    └─ client/          # index.html + client.js (v1 browser client, served by host)
```

Why a workspace of small crates? The private-API blast radius is confined to `virtual-display`; each crate is independently unit-testable; the media crates (macOS-only) are cleanly separated from the cross-platform `transport / core`, which the future Windows native client reuses.

Crate	Platform	Touches private API?	Reused by native client?
core	all	no	✓
virtual-display	macOS	yes	no (host only)
capture	macOS	no	no (host only)
encoder	macOS	no	no (host only)
transport	all	no	✓
input	macOS	no	no

7. Concurrency & Threading Model

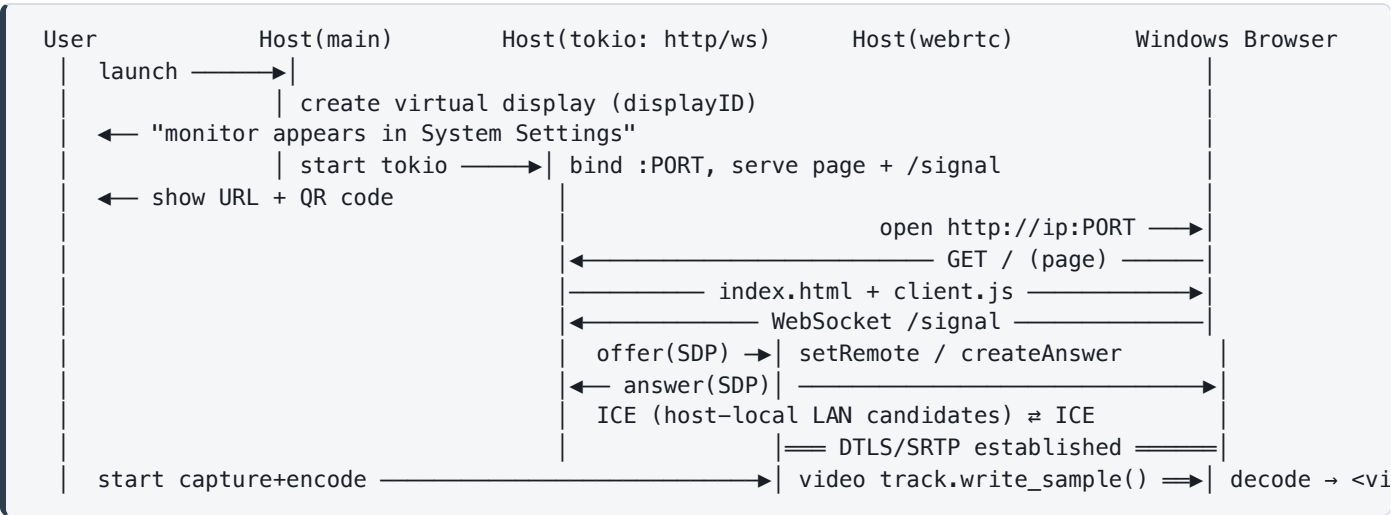
The trickiest systems detail. macOS frameworks impose **thread affinity**; Rust async imposes its own. The design keeps a strict separation between the **main run-loop world** (Apple frameworks) and the **tokio world** (networking), joined by lock-free channels.



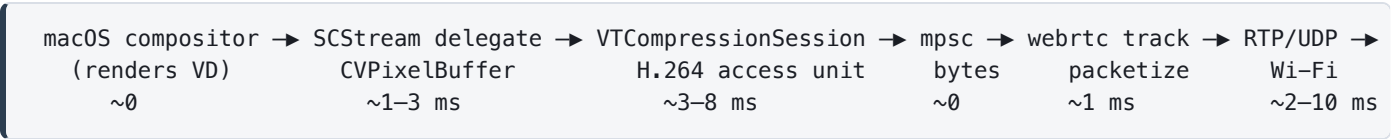
- **Main thread** owns everything AppKit/CoreGraphics/ScreenCaptureKit and runs the run loop.
- **ScreenCaptureKit** calls the frame delegate on its own dispatch queue; we encode there (or hop to a dedicated encode thread) and push bytes onto an `mpsc`.
- **Tokio** runs on background threads (spawned before the run loop takes over the main thread); it drains the channel and writes to the WebRTC track. Networking never blocks capture.
- **Back-pressure**: a bounded channel; if the network stalls, drop the oldest frame and ask the encoder for a keyframe rather than queueing latency.

8. Sequence Diagrams

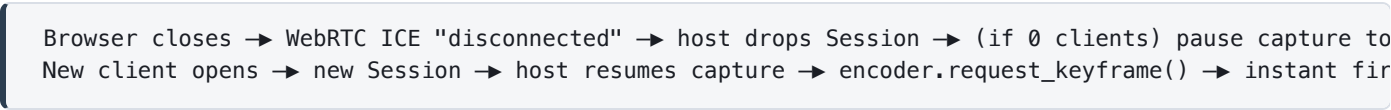
8.1 Startup & connection handshake



8.2 Steady-state frame pipeline (per frame, ~16 ms @ 60 fps)



8.3 Client disconnect / reconnect



The shared vocabulary lives in `core` . Everything else depends on these.

[illegible]

10. Performance & Latency Budget

At 60 fps each frame owns a **16.6 ms** slot; at 30 fps, 33 ms. "Glass-to-glass" latency (a change on the Mac appearing on the Windows screen) is the sum of pipeline stages plus buffering. Realistic targets on a decent 5 GHz Wi-Fi link:

Stage	Typical	Levers to reduce it
Capture (ScreenCaptureKit)	1–3 ms	NV12 output, cap FPS, dirty-region hints
Encode (VideoToolbox HW)	3–8 ms	RealTime, no B-frames, no reordering, sane bitrate
Packetize + queue	<1 ms + queue	bounded channel, drop-oldest back-pressure
Network (Wi-Fi LAN)	2–10 ms	5 GHz / Wi-Fi 6, low channel congestion, wired > wireless
Jitter buffer + decode (browser)	5–16 ms	low-latency playback, small jitter buffer
Total (glass-to-glass)	~30–60 ms	Ethernet/USB-tether pushes toward the low end

Bandwidth: 1080p60 at H.264 real-time is roughly **8–20 Mbit/s** depending on motion. Static desktops compress to almost nothing; video/scrolling spikes it. WebRTC's congestion control adapts bitrate automatically, so wire that signal to `encoder.set_bitrate()` .

- **Idle efficiency:** ScreenCaptureKit only emits frames on change, so a still desktop costs ~0.
- **Motion smoothness** matters more than resolution for perceived quality; prefer 60 fps at a tuned bitrate over 4K at low fps.
- **HEVC** cuts bitrate ~30–40% but only for clients that can decode it, so keep H.264 as the fallback.

11. Security, Permissions & Privacy

- **Screen Recording permission** (TCC): required by ScreenCaptureKit. macOS prompts on first capture; the app appears under **System Settings → Privacy & Security → Screen Recording**.
- **Accessibility permission**: required only for phase-2 input injection (`CGEvent`).
- **Network exposure**: the host opens an HTTP/WebSocket port on the LAN. Anyone on the same network could connect. Mitigate with a **shared-secret token** (`Hello{token}`) shown in the tray UI and encoded in the QR/URL; reject sessions without it.
- **Transport encryption**: WebRTC media is **always encrypted** (DTLS-SRTP). The *signaling* WebSocket is plain `ws://` on the LAN by default; optionally upgrade to `wss://` with a self-signed cert if the threat model warrants.
- **No data leaves the LAN**: no STUN/TURN, no cloud, a strong privacy story vs. commercial tools.
- **Code signing / notarization**: sign the app (Developer ID) so Gatekeeper allows it; it cannot be notarized for the Mac App Store because of the private API, but Developer-ID distribution is fine.

12. Dependencies (Crates) & Build

Version note: the Apple-framework binding ecosystem moves fast. Treat the versions below as "known-good families," and run `cargo add` to pin the current release at build time. Prefer `objc2`'s official framework crates where they exist.

Concern	Recommended crate(s)	Role
Obj-C runtime / Foundation	<code>objc2</code> , <code>objc2-foundation</code> , <code>objc2-app-kit</code>	Call Apple frameworks; runtime class lookup for the private <code>CGVirtualDisplay</code> .
Obj-C blocks / dispatch	<code>block2</code> , <code>dispatch2</code> (or <code>libdispatch FFI</code>)	terminationHandler block; dispatch queues for descriptor + <code>SCStream</code> .
Capture + media types	<code>objc2-screen-capture-kit</code> , <code>objc2-core-media</code> , <code>objc2-core-video</code> , or <code>cidre</code>	ScreenCaptureKit + <code>CVPixelBuffer/CMSampleBuffer</code> . <code>cidre</code> also wraps <code>VideoToolbox</code> richly.
Video encode	<code>cidre</code> (<code>VideoToolbox</code>) or raw extern "C" FFI	<code>VTCompressionSession</code> is a C API; <code>cidre</code> gives the smoothest Rust ergonomics.
WebRTC	<code>webrtc</code> (<code>webrtc-rs</code>)	Pure-Rust peer connection, tracks, DataChannels, RTP.
Async + HTTP/WS signaling	<code>tokio</code> , <code>axum</code> , <code>tokio-tungstenite</code>	Runtime, static page server, WebSocket signaling on one port.
Serialization	<code>serde</code> , <code>serde_json</code>	Signaling + config messages.
Tray UI + QR	<code>tray-icon</code> , <code>muda</code> , <code>qrcode</code>	Menu-bar app, connect URL, QR code.
Native client (phase 2)	<code>winit</code> , <code>wgpu</code> , <code>webrtc</code> , decoder via <code>windows / ffmpeg-next</code>	Windows full-screen renderer + hardware decode.

12.1 Toolchain & build

```
# prerequisites: Xcode command-line tools (for the macOS SDK), stable Rust
rustup toolchain install stable
# build the host (macOS)
cargo build -p host --release
# run (grants Screen Recording prompt on first launch)
cargo run -p host
```

Linking the private API: because we resolve `CGVirtualDisplay` classes with `AnyClass::get("...")` at runtime, there is *no* link-time symbol to satisfy, since `CoreGraphics` is already loaded, so no special linker flags are needed. Screen Recording/Accessibility entitlements are handled via the app bundle's `Info.plist` usage strings and TCC prompts.

13. Risks & Mitigations

Risk	Impact	Mitigation
Private API breaks on a macOS update	Virtual display fails to create	Isolated in one crate; mirror DeskPad/BetterDummy (actively maintained); pin/test per macOS version; fall back to an \$8 HDMI dummy-plug path (100% public APIs).
objc2 / cidre API churn	Build breaks on upgrade	Pin exact versions in <code>Cargo.lock</code> ; upgrade deliberately.
VideoToolbox FFI complexity	Slow encoder milestone	Use <code>cidre</code> 's VT wrappers instead of hand-rolled FFI; validate with a local decode loop first.
Wi-Fi latency/jitter	Laggy feel	5 GHz/Wi-Fi 6; drop-oldest back-pressure; offer wired/USB-tether path; adaptive bitrate.
Browser codec quirks	Black video on some clients	Default to H.264 Constrained Baseline, Annex-B, in-band SPS/PPS; HEVC only when negotiated.
Run-loop / thread-affinity bugs	Crashes, no frames	Strict main-thread rule for Apple APIs; channels to tokio; never call AppKit off-main.
Ghost displays after crash	Leftover virtual monitors	RAII <code>Drop</code> + a termination handler; on launch, reclaim/cleanup stale displays.
Input coordinate mapping (phase 2)	Cursor offset/misclick	Map through the display's global origin + HiDPI scale; test across arrangements.

14. Roadmap & Milestones

Sequenced to **de-risk early** and produce a visible demo at each step. Estimates assume one developer comfortable with Rust, learning the macOS specifics.

#	Milestone	Deliverable / demo	Rough effort
M0	Virtual-display spike	Run a binary → a fake monitor appears in System Settings; drops cleanly on exit.	1–3 days
M1	Capture to disk	Capture the virtual display via ScreenCaptureKit; save frames/PNG to prove the source works.	2–4 days
M2	Encode + local verify	VideoToolbox → H.264; decode locally (or write .h264/.mp4) to confirm a valid low-latency stream.	3–5 days
M3	WebRTC to browser (mirror)	Stream the <i>main</i> display to a browser full-screen; the whole network path proven end-to-end.	4–7 days
M4	Extended display E2E	Point capture at the virtual display → drag a window over → it appears on Windows. v1 done.	2–4 days
M5	Interactive input	DataChannel input → CGEvent injection; mouse/keyboard work on the second screen.	~1 week
M6	Polish	Tray UI, QR/URL, auth token, resolution picker, auto-reconnect, multi-client, HEVC.	ongoing
M7	Native Rust client	<code>winit + wgpu</code> Windows app with HW decode for lowest latency.	1–2 weeks

Fastest path to "it works": M0 → M1 → M2 → M3 → M4. Note that M3 uses the *real* main display so you can prove the streaming half **before** depending on the private virtual-display API, so the two halves are validated independently, then joined at M4.

15. Testing & Verification Strategy

- **Unit tests** for pure logic: config parsing, signaling message (de)serialization, coordinate mapping math, back-pressure/queue behavior.
- **Encoder round-trip test:** feed a synthetic `CVPixelBuffer` (known pattern) → encode → decode → assert similarity (PSNR threshold). Catches format/Annex-B/SPS-PPS regressions.
- **Capture smoke test:** create a virtual display, render a known color, capture one frame, assert pixels.
- **Integration (headless):** host + a headless WebRTC client (webrtc-rs on the test side) negotiate and receive N frames, which verifies the full signaling + media path in CI without a browser.
- **Manual latency measurement:** display a millisecond timer on the Mac's virtual screen, photograph both screens together, subtract: the classic glass-to-glass method.
- **CI:** build + unit/integration tests on a macOS runner; the private-API creation test is gated to run only on self-hosted/interactive machines where TCC prompts can be pre-authorized.
- **Compatibility matrix:** test the browser client on Chrome, Edge, and Firefox on Windows; record the working codec/profile per browser.

Appendix A: Reverse-engineered CGVirtualDisplay interface

Do not transcribe from memory. Copy the current header from a maintained reference (DeskPad or BetterDummy) and cross-check the two, since field signatures have drifted across macOS versions. This appendix shows the *shape* so you recognize it, not an authoritative header.

```
// Private classes living inside CoreGraphics.framework (not in the SDK).
@interface CGVirtualDisplayMode : NSObject
- (instancetype)initWithWidth:(uint32_t)w height:(uint32_t)h refreshRate:(double)r;
@property(readonly) uint32_t width, height; @property(readonly) double refreshRate;
@end

@interface CGVirtualDisplayDescriptor : NSObject
@property(retain) dispatch_queue_t queue;
@property(copy) NSString *name;
@property uint32_t maxPixelsWide, maxPixelsHigh;
@property CGSize sizeInMillimeters; // physical size → DPI
@property uint32_t vendorID, productID, serialNum;
@property(copy) void (^terminationHandler)(id, id); // Obj-C block
@end

@interface CGVirtualDisplaySettings : NSObject
@property(retain) NSArray<CGVirtualDisplayMode *> *modes;
@property uint32_t hiDPI; // 1 = Retina backing
@property uint32_t rotation;
@end

@interface CGVirtualDisplay : NSObject
- (instancetype)initWithDescriptor:(CGVirtualDisplayDescriptor *)d;
- (BOOL)applySettings:(CGVirtualDisplaySettings *)s;
@property(readonly) CGDirectDisplayID displayID; // ← feed to ScreenCaptureKit
@end
```

Creation recipe (from the Rust `virtual-display` crate, conceptually): look up each class with `AnyClass::get(c"CGVirtualDisplay")` → build descriptor (name, dimensions, size_mm, IDs, main queue) → `alloc/initWithDescriptor:` → build settings (one mode, hiDPI) → `applySettings:` → read `displayID`. Retain the object for the session; drop it to remove the monitor.

Appendix B: References

Reference implementations to read / adapt

- **DeskPad**: Swift virtual monitor, MIT. Gold-standard for the `CGVirtualDisplay` creation code.
github.com/Stengo/DeskPad
- **BetterDummy**: Swift virtual display, MIT. Second independent reference for the private API.
github.com/waydabber/BetterDummy
- **BetterDisplay**: the productized, feature-rich descendant. github.com/waydabber/BetterDisplay
- **SimpleDisplay**: a current (macOS 14+) menu-bar app doing the same `CGVirtualDisplay` trick. Useful as a third, recently-maintained cross-check on the header. **GPL-3.0**, see the licence note below. simpledisplay.app
- **Deskreen**: Electron/WebRTC browser-as-second-screen; the streaming/architecture model.
github.com/pavlobu/deskreen

Licence hygiene on the virtual-display crate: DeskPad and BetterDummy are **MIT**; SimpleDisplay is **GPL-3.0**. GPL-3.0 is copyleft, so lifting its code into `crates/virtual-display` would oblige Annex to ship under GPL-3.0 as well. Read SimpleDisplay to *understand* the API and to confirm what still works on current macOS, but transcribe the header and the creation recipe from the MIT projects. None of these tools solve Half 2: SimpleDisplay's own docs tell you to reach the virtual display over VNC, Screen Sharing, or Parsec, which is precisely the gap Annex closes.

Rust libraries

- **objc2** family: Apple framework bindings + runtime. github.com/madsmtm/objc2
- **cidre**: comprehensive Rust bindings for ScreenCaptureKit, CoreMedia, VideoToolbox. github.com/yury/cidre
- **webrtc-rs**: pure-Rust WebRTC. github.com/webrtc-rs/webrtc
- **tokio / axum, winit / wgpu, tray-icon / muda**.

Apple documentation

- ScreenCaptureKit: developer.apple.com/documentation/screencapturekit
- VideoToolbox (VTCompressionSession): developer.apple.com/documentation/videotoolbox
- Quartz Display Services / CGEvent: developer.apple.com/documentation/coregraphics

Appendix C: Glossary

Term	Meaning
CGVirtualDisplay	Private CoreGraphics class that creates a software "monitor" macOS treats as real.
CGDirectDisplayID	Numeric identifier macOS assigns each display; the handle passed between capture and injection.
ScreenCaptureKit (SCK)	Apple's modern, GPU-accelerated screen-capture framework (macOS 12.3+).
CVPixelBuffer	A GPU-friendly image buffer; the frame format passed capture → encoder.
VideoToolbox (VT)	Apple's hardware video encode/decode framework.
NAL unit / Annex-B	H.264 packaging; Annex-B is the start-code stream WebRTC's payload expects.
WebRTC	Real-time media/data transport with built-in encryption, NACK, and congestion control.
SDP / ICE	WebRTC's session description and connectivity-establishment protocols (exchanged via signaling).
DataChannel	WebRTC's arbitrary-data channel; carries phase-2 input events.
Signaling	The out-of-band exchange (our WebSocket) that bootstraps a WebRTC connection.
TCC	macOS "Transparency, Consent & Control": the permission system behind Screen Recording/Accessibility prompts.
RAII / Drop	Rust pattern where releasing a value frees its resource; here, removing the virtual display.
Glass-to-glass latency	Time from a change on the Mac to it appearing on the Windows screen.
HiDPI	Retina-style rendering where the backing store has 2× the logical pixels.

Annex: Architecture & Design v1.0 · Prepared 24 July 2026 · This document describes a design intended for personal/self-hosted use. The virtual-display technique relies on private Apple APIs and is not eligible for Mac App Store distribution.